

EDGAR COUNTY AP, IL, USA

WMO: 722172

Lat: 39.700N

Lon: 87.669W

Elev: 654

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	6.7	-5.6	4.2	2.1	0.0	5.6	7.0	29.7	33.1	26.9	34.6	5.2	310	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	91.1	75.0	89.1	74.9	86.7	73.9	79.6	87.0	77.7	85.2	76.1	83.4	8.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	146.0	84.9	75.4	136.5	82.2	73.8	129.2	80.3	43.6	87.2	41.6	84.9	40.0	83.5	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.5	18.8	-6.0	94.0	9.4	3.2	-12.8	96.3	-18.3	98.2	-23.6	100.0	-30.4	102.4		
			-6.5	82.3	9.2	1.5	-13.2	83.4	-18.6	84.3	-23.8	85.2	-30.5	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	27.9	31.1	42.4	53.5	64.5	72.2	73.5	72.3	67.1	55.3	42.7	33.1	(6)
(7)		DBStd	18.81	12.37	11.90	11.38	9.50	8.90	5.71	5.54	5.49	7.05	9.47	9.63	10.65	(7)
(8)		HDD50	2437	690	535	295	73	6	0	0	0	49	258	531		(8)
(9)		HDD65	5355	1151	949	705	357	124	10	4	8	56	331	671	989	(9)
(10)		CDD50	3557	3	7	59	178	455	666	729	690	514	212	37	7	(10)
(11)		CDD65	999	0	0	4	12	108	226	267	234	119	29	0	0	(11)
(12)		CDH74	9363	0	0	31	159	1004	2028	2483	2161	1236	259	2	0	(12)
(13)		CDH80	3150	0	0	3	18	304	694	891	735	431	74	0	0	(13)
(14)	Wind	WSAvg	7.3	9.9	9.1	9.1	9.6	7.9	5.8	4.0	3.8	4.3	7.1	8.4	8.7	(14)
(15)	Precipitation	PrecAvg	40.8	2.4	2.1	3.2	4.0	4.4	3.9	4.4	3.9	3.1	2.8	3.5	3.1	(15)
(16)		PrecMax	57.2	8.0	5.5	7.5	9.0	12.5	12.7	9.7	8.2	9.5	10.0	10.5	9.1	(16)
(17)		PrecMin	30.6	0.1	0.4	0.7	0.3	0.7	0.5	0.6	0.4	0.2	0.5	0.4	0.6	(17)
(18)		PrecStd	6.7	1.7	1.3	1.7	2.0	2.6	2.4	2.6	2.1	2.4	1.8	2.2	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	65.3	78.6	82.2	89.6	92.3	97.6	93.0	92.8	87.7	72.3	63.1	(19)
(20)		MCWB	57.6	57.8	64.9	62.8	70.4	73.7	76.3	75.9	73.4	71.6	59.2	59.0		(20)
(21)		2%	DB	55.0	59.3	71.5	78.9	86.7	89.8	90.8	89.8	89.6	81.2	67.1	57.5	(21)
(22)		MCWB	52.3	55.9	60.9	62.5	70.3	74.6	77.2	75.9	72.4	66.7	58.8	54.9		(22)
(23)		5%	DB	50.3	53.7	65.0	74.7	83.7	87.3	88.0	86.9	85.1	76.5	62.3	52.9	(23)
(24)		MCWB	47.9	49.8	57.8	61.0	68.5	73.9	77.2	75.2	71.1	63.9	56.2	50.7		(24)
(25)		10%	DB	44.7	48.3	60.9	70.3	80.4	84.1	84.8	84.3	81.3	71.9	57.2	47.9	(25)
(26)		MCWB	42.1	44.8	54.8	59.2	67.4	71.8	75.1	73.8	67.3	62.5	51.0	45.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	60.3	65.8	68.0	75.4	79.7	82.3	81.4	77.5	73.1	64.9	60.2	(27)
(28)		MCDWB	60.5	63.7	76.8	76.4	84.4	87.8	89.5	88.5	87.6	83.8	69.5	62.3		(28)
(29)		2%	WB	53.7	55.7	62.4	65.1	72.7	77.4	80.1	79.1	74.8	70.2	60.4	55.2	(29)
(30)		MCDWB	54.9	58.4	69.5	72.9	82.3	85.6	87.2	86.8	83.8	77.8	64.1	56.8		(30)
(31)		5%	WB	48.2	50.9	59.5	63.1	70.9	75.6	78.2	77.2	72.8	67.4	56.6	50.4	(31)
(32)		MCDWB	50.0	53.8	64.5	71.0	79.7	83.6	85.1	84.3	81.2	72.3	61.5	52.2		(32)
(33)		10%	WB	42.0	44.6	55.5	60.9	69.0	73.9	76.4	75.4	70.9	64.0	52.6	45.6	(33)
(34)		MCDWB	44.5	47.8	60.0	68.4	77.3	81.2	83.3	81.5	78.1	69.5	56.9	47.6		(34)
(35)	Mean Daily Temperature Range	MDBR	14.9	16.1	18.0	21.0	20.1	19.9	19.7	21.5	24.1	21.3	17.7	13.6		(35)
(36)		5% DB	MCDDBR	20.0	22.2	24.0	26.8	23.8	22.6	22.0	23.0	27.2	25.4	23.0	18.8	(36)
(37)		MCWBR	18.0	18.8	15.6	14.2	10.4	10.6	10.8	11.3	11.7	13.1	15.8	16.7		(37)
(38)		MCDDBR	19.4	21.8	21.6	22.1	20.7	18.6	19.4	19.5	22.2	21.1	20.2	18.2		(38)
(39)		MCWBR	18.3	19.3	14.6	14.3	10.8	10.1	11.2	11.0	10.3	11.9	16.0	17.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.316	0.356	0.405	0.441	0.445	0.464	0.473	0.427	0.364	0.317	0.299		(40)
(41)		taud	2.440	2.397	2.362	2.227	2.178	2.208	2.146	2.109	2.218	2.396	2.482	2.473		(41)
(42)		Ebn at Noon	275	287	285	276	267	266	259	253	257	265	268	268		(42)
(43)		Edh at Noon	25	31	36	43	47	45	48	48	41	30	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	566	836	1164	1509	1722	1977	1931	1747	1463	1021	681	479		(44)
(45)		RadStd	64	89	95	104	153	119	115	108	104	102	70	56		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page